MATH0111 Frontiers in Computational Mathematics

Year:	2026–2027
Code:	MATH0111
Level:	7 (UG and PG)
Normal student group(s):	UG: Year 4 or Year 3 Mathematics degrees PG: MSc Mathematics
Value:	15 credits (= 7.5 ECTS credits)
Term:	2
Assessment:	85% examination, 15% coursework
Normal Pre-requisites:	A 2nd or 3rd year UG module in computational mathematics e.g. (for UCL students) MATH0058 or MATH0033
Lecturer:	Prof E Burman, Prof T Betcke, Dr M Jensen

Course Description and Objectives

This module will introduce state-of-the-art numerical methods for solving complex problems in mathematics, science, and technology. Topics range from advances in the theory of numerical analysis to mathematical challenges arising in high-performance computing. The course aims to deepen understanding of numerical techniques in current research and real-world applications through topical academic texts and research papers.
The module is split into three blocks, each introducing a separate, current development in Computational Mathematics. Typically, each block is taught by a different lecturer to take full advantage of the expertise in Computational Mathematics at UCL.

Recommended Texts

Course materials are provided through a combination of lecture notes, UCL-licensed texts, and open-access resources. Reading lists are regularly updated and published on ReadingLists@UCL. The following are representative examples:

https://link.springer.com/book/10.1007/978-3-030-56341-7

https://link.springer.com/book/10.1007/978-3-030-56923-5

https://link.springer.com/book/10.1007/978-3-030-57348-5

https://doi.org/10.1017/S0962492917000071

https://doi.org/10.1017/S0962492920000021

Detailed Syllabus

The focus of the topics is subject to change from year to year, with the inclusion of guest lecturers and new research developments, but examples may include:
1. Numerical Analysis 2. Numerical Linear Algebra 3. Numerical Solutions of Differential Equations 4. Optimisation and Nonlinear Equations 5. Data-driven Scientific Computing 6. Computational Mathematics for Real-World Problems